

Q1 - Chap 1 (pg 1-8), (pg 20-23)

Ref:

Date:

(a) part

a)

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Case 1

 $A = \{x \mid x \text{ is a prime number}\}$ $B = \{x \mid x \text{ is a positive integer and } x^2 \leq 25\}$ $C = \{x \mid x \text{ is a multiple of 5}\}$

i) List all elements of sets A, B and C

 $A = \{2, 3, 5, 7\}$ $B = \{1, 2, 3, 4, 5\}$ $C = \{5, 10\}$ ii) $(A \cap B) \oplus C \approx (\{2, 3, 5, 7\} \cap \{1, 2, 3, 4, 5\}) \oplus C$ Find $\Rightarrow \{5, 7\} \oplus \{5, 10\} = \{(5, 7) - (5, 10)\} \cup \{(5, 10) - (5, 7)\}$
 $= \{7\} \cup \{10\} = \{7, 10\}$ 2) $A - C$ $\approx \{2, 3, 5, 7\} - \{5, 10\}$ $\approx \{2, 3, 7\}$ 3) $\overline{B} \cup C$ $\approx \{1, 2, 3, 4\} \cup \{5, 10\}$ $= \{1, 2, 3, 4, 5, 10\} \approx \{6, 7, 8, 9\}$

Case 2

 $A = \{x \mid x \text{ is an odd number}\} \mid B = \{x \mid x \text{ is multiple of 3}\}$ $C = \{x \mid x \text{ is a factor of 18}\}$

i) List all elements of sets A, B, and C

 $A =$ $B =$ $C =$ ii) $A \cap B \approx$ Find $\Rightarrow$ i) $A - B \approx$ $B - A \approx$ $A \oplus B \approx$ $\overline{A \cup C} \approx$

(b) part

(b) Case 2 // Given $a = 2020$ and $b = 92$ i) Use the Euclidean algorithm to find the ~~the~~ GCD of a and b .

$$\text{GCD}(2020, 92) = 4$$

iii) compute the Least Common Multiple (LCM)

$$\text{LCM} = \frac{a \times b}{\text{GCD}(a, b)} \approx \frac{2020 \times 92}{4} = 46460$$

$$2020 = 92(21) + 88$$

$$92 = 88(1) + 4$$

$$88 = 4(22) + 0$$

use
back~~ii)~~ii) GCD is the form of $sa + tb$, where $s, t \in \mathbb{Z}$.

$$4 = 92 - 88(1)$$

$$4 = 92 - [2020 - 92(21)](1)$$

$$= 92(1) - 2020(1) + 92(21)$$

$$= 92(22) - 2020(1)$$

$$= 2020(-1) + 92(22)$$

$$s = -1 \quad t = 22$$

 $s(a + tb)$

so 转换

$$88 = 2020 - 92(21)$$

$$4 = 92 - 88(1)$$

b) Case 1 // Given $a = 2921$ and $b = 460$ i) Find GCD of a and b

$$\text{GCD}(2921, 460) = 23$$

iii) Find LCM

$$\text{LCM} = \frac{2921 \times 460}{23} \approx 58420$$

$$2921 = 460(6) + 161$$

$$460 = 161(2) + 138$$

$$161 = 138(1) + 23$$

$$138 = 23(6) + 0$$

~~ii)~~ ii) $sa + tb$, where $s, t \in \mathbb{Z}$

$$23 = 161 - 138(1)$$

$$23 = 161 - [460 - 161(2)]$$

$$= 161(1) - 460(1) + 161(2)$$

$$= 161(3) - 460(1)$$

$$= [2921 - 460(6)](3) - 460(1)$$

$$= 2921(3) - 460(19)$$

$$= 2921(3) + 460(-19)$$

$$s = 3 \quad t = -19$$

$$161 = 2921 - 460(6)$$

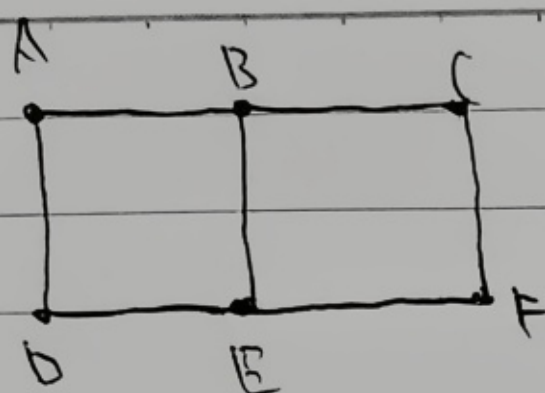
$$138 = 460 - 161(2)$$

$$23 = 161 - 138(1)$$

转换

(C) part

(C)



i) Write the degree of each vertex of the graph.

Vertex	A	B	C	D	E	F
degree	2	3	2	2	3	2

ii) Identify Euler circuit. If it does, write down a Euler circuit by listing the vertices in the order they are visited, starting and ending at vertex A. If it does not, explain why.

Euler circuit $\approx$ All vertex of even degree, there can be Euler circuit
 $\approx$ a vertex of odd degree, there can be no Euler circuit

Ans: No, ~~the graph does not contain an Euler circuit~~. This is because vertices B and E have an odd degree of 3. // Yes, — every vertex has even degree.

iii) Identify Euler path. If it does, write down a Euler path starting at vertex B. If it does not, explain why.

Euler path $\approx$ When has exactly two vertices of odd degree, there is Euler path
 $\approx$ When has more than two vertices of odd degree, then no Euler path

Ans: Yes, This is because ^{graph} has exactly two vertices of odd degree.
 One Euler path is: $B \rightarrow A \rightarrow D \rightarrow E \rightarrow B \rightarrow C \rightarrow F \rightarrow E$

// No, — graph has more than two vertex of odd degree.
 , does not have any vertex of odd degree.

(iv) Identify, ... Hamiltonian circuit. If it does, write down a Hamiltonian circuit ... at vertex C. If it does not, explain why. (每个点走到完)

Hamiltonian circuit $\Rightarrow$ 从 ? 回到 ? , 每个 vertex 都要经过一次, 才算

Ans: Yes. One such circuit starting and ending at C is: $C \rightarrow F \rightarrow E \rightarrow D \rightarrow A \rightarrow B \rightarrow C$

(No, skip the ans) // write "No" only.

Extra $\rightarrow$ Chap 1 (pg 24-28)

// same logic like programming

1) $A \vee B$

$$1 \vee 1 = 1$$

OR 全部都是对,

OR

$$\approx 1 \vee 0 = 1$$

除了 $0 \vee 0 = 0$

$$0 \vee 1 = 1$$

$$0 \vee 0 = 0$$

2) $A \wedge B$

$$1 \wedge 1 = 1$$

AND 全部都是错,

AND

$$\approx 1 \wedge 0 = 0$$

除了 $1 \wedge 1 = 1$

$$0 \wedge 1 = 0$$

$$0 \wedge 0 = 0$$

3) $A \oplus B$

异或

$\approx$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$\oplus$

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\approx$ 直接丢

(a) part

(a) Let $A = \{a, b, c, d\}$ // $R = \{(a, a), (a, c), (a, d), (b, b), (b, d), (c, b), (c, d), (d, c), (d, d)\}$

i) Identify the domain and range of the relation R

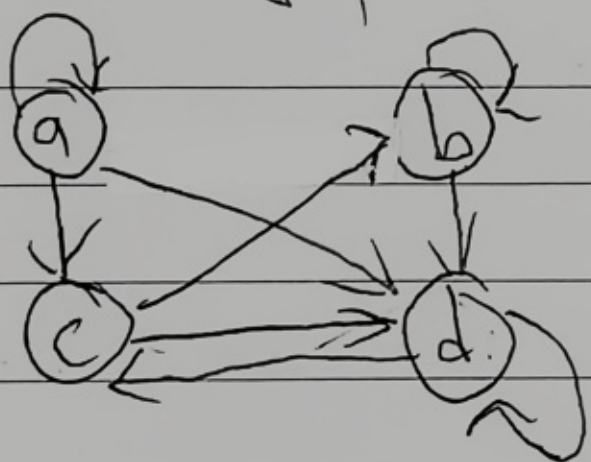
Domain = $\{a, b, c, d\}$ // Range = $\{a, b, c, d\}$

ii) Construct the matrix representation of R

Matrix =

	a	b	c	d
a	1	0	1	1
b	0	1	0	1
c	0	1	0	1
d	0	0	1	1

iii) Draw the digraph of the relation R



iv) Compute the in-degree and out-degree of each vertex

Vertex	a	b	c	d	Σ
In	1	2	2	3	8
Out	3	2	2	2	9

TODO

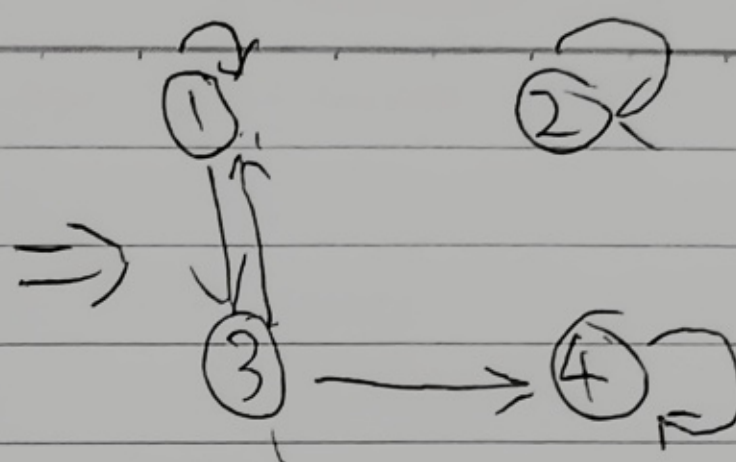
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(b) part

b) $A = \{1, 2, 3, 4\}$

$$M_{123} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$



i)

1) Reflexive (

Ans:

2) Irreflexive (

Ans:

3) Symmetric (

Ans:

4) Asymmetric (

Ans:

5) Antisymmetric (

Ans:

6) Transitive (

Ans:

($\nexists$ Reflexive, symmetric, transitive)

i) Identity, whether R is an equivalence relation on A , Justify your answer
 R is not reflexive since $(3, 3) \notin R$

25

$\therefore R$ is not an equivalence relation

ii) ~~Apply Warshall's algorithm to obtain the matrix of the transitive closure of R .~~

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Next $\longrightarrow$

Q22

look back ln (pg 94-97)

Q22 (cont.)

Ref:

Date:

(b)(iii) (cont.)

(iii) part

(iii) Apply Warshall's algorithm to obtain the matrix of the transitive closure R

$$W_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow$ 找 $n=4$ $\leftarrow \{1, 2, 3, 4\}$
 $W_0 \rightarrow W_4$ $\leftarrow$ (看 1 的部分)

组合起来还有重复的

W_0 (First row, first col)

$$\Rightarrow W_0 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Col 0 = 1, 3 Add

Row 0 = 1, 3

Add = (3, 3)

+1 $\downarrow$

W_1 (Col 2 - Row 2)

$$\Rightarrow W_1 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$C_1 = 2$

$R_1 = 0$

Add = ()

W_2 (Col 3 - Row 3)

$$\Rightarrow W_2 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$C_2 = 3$

$R_2 = 1, 4$

Add = (1, 4)

W_3 (Col 4 - Row 4)

$$\Rightarrow W_3 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$C_3 = 3, 4$

$R_3 = 4$

Add = ((3, 4))

全部

add 进去

原来的

add

$$W_4 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\Leftarrow$ Add (3, 3), (1, 4)

(iv) part)

(iv)

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(iv) Find the matrix of the reflexive closure of R# Reflexive closure of R = $R \cup \Delta \leftarrow (a,a) \in R$ // 全部都要 loopIn case: $\{(1,1), (2,2), (4,4)\}$ / 1, 3 (3,3)

Ans: Add (3,3)

$$M_{\text{reflexive}} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(v) Find the matrix of the ~~reflexive~~ symmetric closure of R# Symmetric closure of R = $R \cup R^{-1}$ // 全部调换过来
/ 反转全部In case: $\{(1,1), (1,3), (2,2), (3,1), (3,4), (4,4)\}$

$$R^{-1} = \{(1,1), (3,1), (2,2), (1,3), (4,3), (4,4)\}$$

$$M_{\text{symmetric}} = M_R \cup M_{R^{-1}} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$M_R \cup M_{R^{-1}}$$

ADD

add this (4,3)

Q3 -

Ref:

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(a) part

(a) Let $A = \{1, 2, 3, 4\}$

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad M_S = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

(i) $M_{\bar{S}}$

Ans:

$$M_{\bar{S}} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

- = not = $1 \Rightarrow 0$ $0 \Rightarrow 1$ AND $\rightarrow 1 \wedge 1 = 1$ (ii) $M_{R^T \wedge S}$

Ans:

 $M_{R^T \wedge S}$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

 $\uparrow$ $\wedge M_S =$

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

 $R^T = \text{transpose}$ $\{(1,1), (1,4), (2,3), (3,2), (3,3), (4,3)\}$ $\{(1,1), (4,1), (3,2), (2,3), (3,3), (3,4)\}$

Ans =

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

(iii) $M_{R \odot S}$ // $\odot = (\text{行与列的 AND}) \text{ OR 结果}$ $R \odot S \Rightarrow$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Small

$$\begin{aligned} \text{Exp} \Rightarrow \text{Row1, Col1} &= (1 \wedge 0) \vee (0 \wedge 1) \vee (0 \wedge 0) \vee (1 \wedge 0) \\ &= 0 \vee 0 \vee 0 \vee 0 \\ &= 0 \end{aligned}$$

直接算, 除非有时间回来估

$$\text{Ans} = \begin{bmatrix} 1 & \dots & \dots \\ \vdots & \ddots & \ddots \end{bmatrix}$$

Row1, Col2 Row1, Col3 ...

(b) OR (c) part

(b) Consider the $(2, 5)$ encoding function $e: B^2 \rightarrow B^5$

$$e(00) = 00000 = C_0$$

$$e(01) = 01100 = C_1$$

$$e(10) = 10101 = C_2$$

$$e(11) = 11001 = C_3$$

- 1 1 0 1 0 1 = 0 // 1 - 1 0 = 1

$$0 \oplus 0 = 0$$

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$1 \oplus 1 = 0$$

1) $\oplus$

	$C_0 = 00000$	$C_1 = 01100$	$C_2 = 10101$	$C_3 = 11001$
$C_0 = 00000$	00000	01100	10101	11001
$C_1 = 01100$	01100	00000	11001	10101
$C_2 = 10101$	10101	11001	00000	01100
$C_3 = 11001$	11001	10101	01100	00000

(ii) Show that the $(2, 5)$ encoding function is a group code.

Group code $\Rightarrow B^5$ identity $\Rightarrow C_0 = 00000$
 $\Rightarrow$ closure

Let $G = \{00000, 01100, 10101, 11001\}$

- 00000 is the identity of B^5 and it is belong to G .
- Every element is its own inverse

The encoding function is a group code

(iii) Compute the minimum distance of the group code and ... number of error that can be detected.

Minimum Distance - The minimum distance = minimum weight $C_1 = 01100 = 2$ $C_3 = 11001 = 3$
 $\Rightarrow$ Ans: $= 2$ $\hookrightarrow C_2 = 10101 = 3$

Error - Minimum distance - 1 = Error
 $2 - 1 = 1$

Q3 (cont.)

Ln (pg 105-110)

Ref:

Date:

(b) OR (c) part

Case 2 (Oct) (b)

(c) Given $A = \{1, 2, 3, 4, 5\}$. Let $p_1 = (1, 3, 5)$ and $p_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 5 & 2 \end{pmatrix}$ be permutations on the set A

Find $p_1 = (1, 3, 5)$ $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{pmatrix}$

i) Compute p_1^{-1}

Ans = $p_1^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{pmatrix} \approx \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 1 & 4 & 3 \end{pmatrix}$

排列上 p_1^{-1} , $= 3 \rightarrow 1, 2 \rightarrow 2, 5 \rightarrow 3, 4 \rightarrow 4, 1 \rightarrow 5$, 跟着顺序排
 $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

ii) Compute $p_1^{-1} \circ p_2$

Exp: $1 \rightarrow 3 \rightarrow 3 \rightarrow 1$

$p_1^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 1 & 4 & 3 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 5 & 2 \end{pmatrix}$

Ans = $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 5 & 3 & 2 \end{pmatrix}$

Write $(p_1^{-1} \circ p_2)$ as a product of disjoint cycles, and also as a product of transpositions.

$p_1^{-1} \circ p_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 1 & 4 & 3 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 5 & 2 \end{pmatrix}$

看有没有 loop $\approx \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 5 & 3 & 2 \end{pmatrix} = 2 \rightarrow 4 \rightarrow 3 \rightarrow 5 \rightarrow 2$
 $= (2, 4, 3, 5) \approx (b_i, b_r) \circ (b_i, b_{r-1}) \circ \dots$
 $= (2, 5) \circ (2, 3) \circ (2, 4)$

(v) Determine whether permutation $(p_1^{-1} \circ p_2)$ is even or odd. Justify your answer.

$(2, 4, 3, 5)$
 $= (2, 5) \circ (2, 3) \circ (2, 4)$
 $= \text{Number of transpositions} = 3$

Ans: odd permutation // even permutation

Because the cycle is (?) use back contains an even number/odd number of elements

$R = \text{result}$

Q4 -

(a) part

Ref: $P \leftrightarrow Q$
same $= R = T$

Date:

(a) Let $A = (\sim p \vee q) \leftrightarrow (r \rightarrow p)$

// $\wedge = \text{AND}$ // $P \rightarrow Q$
// $\vee = \text{OR}$ // $P=T, Q=F, R=F$

p	q	r	$\sim p$	$\sim p \vee q$	$r \rightarrow p$	$(\sim p \vee q) \leftrightarrow (r \rightarrow p)$
i) p	q	r	$\sim p$	$\sim p \vee q$	$r \rightarrow p$	$(\sim p \vee q) \leftrightarrow (r \rightarrow p)$
T	T	T	F	T	T	T
T	T	F	F	T	T	T
T	F	T	F	F	T	F
T	F	F	F	F	T	F
F	T	T	T	T	F	F
F	T	F	T	T	T	T
F	F	T	T	T	F	F
F	F	F	T	T	T	T

$\rightarrow$ # 全部为 R $\rightarrow T$

(ii) Identify whether the expression A is

- Tautology - All T
- Contradiction - All F
- Contingency - T OR F

Ans: Contingency

T, T, F, F, T, F, F, T

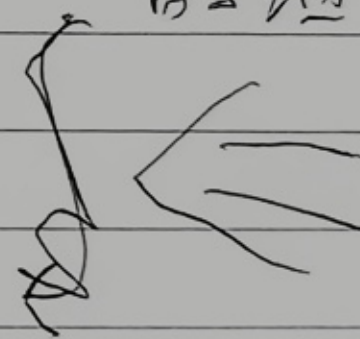
(iii) p DNF # 找 True 的 组合

看 p, q, r 的 组合

Format: $pqr + pqr + \dots$

PDNF

Ans: PDNF of A = $pqr + pq\bar{r} + \bar{p}q\bar{r} + \bar{p}\bar{q}\bar{r}$



(iii) PCNF # R is False 的 组合 // Format: $(p+q+r)(p+q+r)$ // 全部 的 组合

PCNF of A = $(\bar{p}+q+\bar{r})(\bar{p}+q+r)(p+\bar{q}+\bar{r})(p+q+\bar{r})$ // Exp $(p+q+r)$
 $(\bar{p}+\bar{q}+\bar{r})$

(iii)(cont.)

Ref:

Date:

Q
(iii) Write the converse, inverse, contrapositive and negation statements for the following statement:
Extra

"If I exercise regularly, then I stay healthy,"

1) Converse $P \rightarrow Q \Rightarrow Q \rightarrow P$ // stay If..., then ...
Q P

Ans: If I stay healthy, then I exercise regularly

2) Inverse $P \rightarrow Q \Rightarrow \sim P \rightarrow \sim Q$ // stay If..., then ...
 $\sim P$ $\sim Q$

Ans: If I not exercise regularly, then I not stay healthy.

3) Contrapositive $P \rightarrow Q \Rightarrow \sim Q \rightarrow \sim P$ // stay If..., then ...
 $\sim Q$ $\sim P$

Ans: If I not stay healthy, then I not exercise regularly

4) Negation $P \rightarrow Q \Rightarrow P \wedge \sim Q$ // No need If..., then ...
P $\wedge$ $\sim Q$

Ans: I exercise regularly and I not stay healthy.

Q4 (cont.) - Chap 7 (Boolean Algebra)

$f = v = \text{AND}$

Ref: $= 1 = \text{OR}$

Date:

(b)(c)(d) part

(c) Simplify $[(a \vee b)' \vee (a \wedge b')] \wedge b$ to its simplest form using the laws of Boolean algebra

(ii)

$$[(a \vee b)' \vee (a \wedge b')] \wedge b$$

$\frac{16}{17}$ Law

$$= [(a' \wedge b') \vee (a \wedge b')] \wedge b \quad (\text{Distributive laws})$$

$$= [(a' \vee a) \wedge b'] \wedge b$$

$(a' \vee a = 1)$ (complement law)

$$= [1 \wedge b'] \wedge b \Rightarrow \cancel{b' \wedge 1} \quad b' \wedge 1 = b' = b' \wedge b$$

(Identity Law)

$$= b' \wedge b \Rightarrow b' \wedge b = 0 \quad (\text{complement law})$$

$\hookrightarrow P \wedge \sim P = F$

$$= b' \wedge b = 0$$

$$\text{Ans: } [(a \vee b)' \vee (a \wedge b')] \wedge b = 0$$

(ii) Let $f(x, y, z, w) = (x' \wedge y \wedge z \wedge w) \vee (x' \wedge y' \wedge z' \wedge w') \vee (x \wedge y \wedge z \wedge w) \vee (x \wedge y' \wedge z' \wedge w')$

$$f = x'yzw + x'y'z'w' + xyzw + xy'z'w'$$

有到对应的格子, 为 1, 其他的为 0

	z'	z'	z	z	
x'	1	0	0	0	y'
x'	0	0	1	1	y
x	0	0	1	1	x'
x	1	0	0	0	y'
	w'	w	w	w'	

(iv) 不用做, 直接表